

GROUP-INVARIANT BUTSON HADAMARD MATRICES AND A PRIME DIVIDING THE ORDER EXACTLY ONCE

ABSTRACT. We prove that if a prime p divides the order of a finite abelian group G exactly once and is unramified in $\mathbb{Q}(\zeta_h)$, then there is no G -invariant Butson Hadamard matrix $\text{BH}(G, h)$; equivalently, no generalized bent function $G \rightarrow \mathbb{Z}_h$ and, for cyclic G , no perfect sequence of length $|G|$ over μ_h . The proof uses the local ring $\mathbb{F}[T]/(T-1)^p$ and Parseval's identity on a single coset of the Sylow p -subgroup. Consequences: the conjectures of Mow and of Do Duc hold for square-free lengths; the case $m = 1$ of the Kumar–Scholtz–Welch problem on generalized bent functions $\mathbb{Z}_q^m \rightarrow \mathbb{Z}_q$, $q \equiv 2 \pmod{4}$, is settled; the forbidden subgroup of a semi-regular abelian relative difference set is a p -group whenever p divides its index exactly once; and the eleven orders below 1000 left open in Arasu's survey of circulant complex Hadamard matrices are excluded. For the family of perfect p -ary sequences of length pq^e , to which the theorem does not apply when p is odd and $e \geq 2$, we prove nonexistence for all e , using a comparison lemma at the ramified prime p ; for 34 of the 334 such lengths up to 2000 no earlier criterion we know of applies. The main theorem, the comparison lemma (for integer n) and the nonexistence of perfect p -ary sequences of length pq^e are formalized in Lean 4.

1. INTRODUCTION

A *Butson Hadamard matrix* $\text{BH}(n, h)$ [But62] is an $n \times n$ matrix H with entries in the group μ_h of complex h -th roots of unity and $HH^* = nI_n$. For a finite abelian group G , a G -invariant $\text{BH}(G, h)$ has entries $f(g' - g)$ for a function $f : G \rightarrow \mu_h$. For $G = C_n$ these are the circulant $\text{BH}(n, h)$, whose first rows are the *perfect sequences* of length n over μ_h ; equivalent objects are generalized bent functions $G \rightarrow \mathbb{Z}_h$ and, for prime h , semi-regular relative difference sets. Since Turyn [Tur65], nonexistence proofs work with the ideal factorization of the character values of $X = \sum_g f(g)g \in \mathbb{Z}[\zeta_h][G]$, which have absolute value $\sqrt{|G|}$: self-conjugacy, Ma's lemma [Ma85, Pot95], field descent [Sch99], class groups.

1.1. Two conjectures. Let r be the square-free part of n . The following conjecture is attributed to Mow [Mow95] in K.-U. Schmidt's survey [Sch16, Conj. 4.1.1] (we could not consult the text of [Mow95], whose abstract describes a unified construction): *a perfect sequence of length $n > 1$ over μ_h exists if and only if h is divisible by $2\sqrt{rn}$ for $n \equiv 2 \pmod{4}$ and by $\sqrt{rn}$ otherwise.* Do Duc [DD19a, Conj. 3.1]¹ conjectured: *a $\text{BH}(\mathbb{Z}_n, h)$ exists if and only if (i) $\nu_p(h) \geq \lceil \nu_p(n)/2 \rceil$ for every prime $p \mid n$, and (ii) $\nu_2(h) \geq 2$ if $n \equiv 2 \pmod{4}$.* Since $\nu_p(\sqrt{rn}) = \lceil \nu_p(n)/2 \rceil$, the two conjectures are the same statement. Sufficiency follows from the constructions of Mow and of Duc and Schmidt [DDS19, Cor. 2.5], [DD19a, Result 4.1]. Do Duc writes that “we are far from proving Conjecture 3.1” and lists 2687 pairs (n, h) , $n, h \leq 100$, as open [DD19a, Remark 3.13].

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¹We number the results of [DD19a] as in arXiv:1903.07334v2.

1.2. The main theorem.

Theorem 1.1. *Let G be a finite abelian group and p a prime with $p \parallel |G|$. If p is unramified in $\mathbb{Q}(\zeta_h)$, that is, $p \nmid h$, or $p = 2$ and $4 \nmid h$, then there is no $\text{BH}(G, h)$. In particular, if a perfect sequence of length n over μ_h exists, then every prime p with $\nu_p(n) = 1$ divides h , and $4 \mid h$ if $n \equiv 2 \pmod{4}$. [Lean]*

The proof (Section 2) writes $G = H \times C_p$, evaluates only the characters of H , and keeps C_p as a group-ring variable. Modulo a prime above p the group ring of C_p becomes the local ring $\mathbb{F}[T]/(T-1)^p$; this forces all coefficients of the partial evaluation into one prime above p , and Parseval's identity on a single coset of C_p gives a contradiction.

1.3. Consequences. (a) Conditions (i) at every prime p with $\nu_p(n) = 1$ and (ii) are necessary for every abelian group of order n (Corollary 3.3). With the known sufficiency this gives:

Corollary 1.2. *Let n and h be positive integers. If n is square-free, a $\text{BH}(\mathbb{Z}_n, h)$ exists if and only if $n \mid h$ and, for even n , $4 \mid h$. Thus the conjectures of Mow and of Do Duc hold for square-free n ; e.g. a circulant $\text{BH}(2p, h)$, p an odd prime, exists if and only if $4p \mid h$. [Lean-derived]*

Theorem 1.1 excludes 2323 of the 2687 pairs listed in [DD19a, Remark 3.13]; 2063 of them are not excluded by any result known to us, subject to [LCZ25]. (b) For $q \equiv 2 \pmod{4}$ there is no generalized bent function $\mathbb{Z}_q \rightarrow \mathbb{Z}_q$, the case $m = 1$ of the problem of Kumar, Scholtz and Welch [KSW85], treated so far family by family, most recently for $q = 2 \cdot 3^a 7^b$ [YD26] (Corollary 3.4). (c) If D is a semi-regular RDS in an abelian group G relative to N and $p \parallel |G/N|$, then N is a p -group, without any splitting assumption (Corollary 3.1). This contains all nonexistence results of [LSZ23, Sect. 6] and decides eight entries marked “?” in its tables. (d) The eleven orders 260, 340, 442, 468, 520, 580, 680, 754, 820, 884, 890, the only orders ≤ 1000 for which a circulant complex Hadamard matrix had not been excluded according to Arasu's survey [Ara11], are excluded (Corollary 3.5). (e) For perfect quaternary sequences of length $N \equiv 0 \pmod{4}$, $N \leq 5000$, only 3528 and 4900 remain under the criteria we implemented (Appendix A).

(f) *The ramified regime* (Section 4). For the family $\text{BH}(pq^e, p)$, $p \neq q$ primes, Theorem 1.1 applies only if $e = 1$ or $p = 2$. Theorem 4.1 shows that there is no circulant $\text{BH}(pq^e, p)$ for all $e \geq 1$. Earlier results cover the case that q is self-conjugate modulo p [DD19a, Thm. 3.3] and further subfamilies; of the 334 lengths $pq^e \leq 2000$ with p odd and $e \geq 2$, 34 are not covered by the earlier criteria we implemented, among them 56 and 99 from Yayla's table [Yay16]. The tool is a comparison lemma at the ramified prime p (Lemma 4.10), proved by expanding in the uniformizer $\zeta_p - 1$: the digits are polynomials in the exponent of bounded degree, and comparing degrees with the p points of $\mathbb{F}_p$ gives the claim.

The proofs use no result of the (not peer-reviewed) OpenAI preprint on the circulant Hadamard conjecture [Ope26], in whose Lean library our formalization is placed.

Conventions. C_N is the cyclic group of order N , $\zeta_t = e^{2\pi i/t}$, $\nu_p = v_p$ the p -adic valuation and $v_{\mathfrak{P}}$ the valuation at a prime ideal $\mathfrak{P}$. Tags: [Lean]: the statement is an instance of a formally verified Lean theorem (Section 6.1); [Lean-derived]: it follows from a formally verified statement by a short argument that is not formalized; [Comp]: it rests on a finite computation (Section 6.2).

2. PROOF OF THEOREM 1.1

2.1. Notation. For a finite abelian group V and a subring $R \subseteq \mathbb{C}$ closed under complex conjugation, $R[V]$ is the group ring; for $f = \sum_{z \in V} f_z z$ put $f^* = \sum_z \overline{f_z} z^{-1}$ and $\chi(f) = \sum_z f_z \chi(z)$ for $\chi \in \widehat{V}$, so that $\chi(f^*) = \overline{\chi(f)}$. If $V = S \times P$ and $\psi \in \widehat{S}$, the *partial evaluation* $f \mapsto \psi(f) \in R[\psi(S)][P]$ applies ψ to the S -coordinate; it is a ring homomorphism commuting with $*$. The *augmentation* of $F = \sum_z F_z z$ is $F(1) = \sum_z F_z$. For $f : G \rightarrow \mu_h$ and $X_f = \sum_g f(g)g$, the

coefficient of s in $X_f X_f^*$ is $\sum_g f(g+s) \overline{f(g)}$; so f is perfect iff $X_f X_f^* = |G|$, iff $|\chi(X_f)|^2 = |G|$ for all χ (f is a generalized bent function), iff $(f(g' - g))_{g,g'}$ is a $\text{BH}(G, h)$. Thus Theorem 1.1 says: *there is no $X \in \mathbb{Z}[\zeta_h][G]$ with all coefficients in μ_h and $XX^* = |G|$.*

2.2. Setup. Since $p \parallel |G|$, the Sylow p -subgroup is cyclic of order p and is a direct factor: $G = H \times C_p$ with $p \nmid |H|$ and $C_p = \langle T \rangle$. Write

$$(1) \quad X = \sum_{k \in \mathbb{Z}/p} c_k T^k, \quad c_k \in \mathbb{Z}[\zeta_h][H],$$

so that each column c_k has $|H|$ coefficients, all in μ_h . Let $h' = h$ if $p \nmid h$, and $h' = h/2$ if $p = 2$ and $h \equiv 2 \pmod{4}$. Then $p \nmid h'$ and $\mu_h \subseteq \pm \mu_{h'} \subset \mathbb{Z}[\zeta_{h'}]$ (if h' is odd, $-\zeta_{h'}^k$ runs through $\mu_{2h'} \setminus \mu_{h'}$). Let $L = \text{lcm}(h', \exp H)$ and $\mathcal{O} = \mathbb{Z}[\zeta_L]$. Since $p \nmid L$, the prime p is unramified in $\mathcal{O}$; fix a prime $\mathfrak{p}$ of $\mathcal{O}$ above p . For $\chi \in \widehat{H}$ the partial evaluation

$$X_\chi = \sum_{k \in \mathbb{Z}/p} \chi(c_k) T^k \in \mathcal{O}[C_p]$$

satisfies $X_\chi X_\chi^* = |G|$.

2.3. The column lemma.

Lemma 2.1 (Column lemma). *Let $\mathcal{O}$ be the ring of integers of a number field that is closed under complex conjugation and in which p is unramified, let $\mathfrak{p}$ be a prime of $\mathcal{O}$ above p , and let $F \in \mathcal{O}[C_p]$ satisfy $FF^* = n$ with $n \in \mathbb{Z}$, $v_p(n) = 1$. Then exactly one of $F(1)$, $\overline{F(1)}$ lies in $\mathfrak{p}$, and there is $\mathfrak{q} \in \{\mathfrak{p}, \overline{\mathfrak{p}}\}$ such that every coefficient of F lies in $\mathfrak{q}$. In particular $c\bar{c} \in \mathfrak{p}$ for every coefficient c of F .*

The prime $\mathfrak{q}$ may depend on F ; the assertion $c\bar{c} \in \mathfrak{p}$ does not, because $c \in \overline{\mathfrak{p}}$ iff $\bar{c} \in \mathfrak{p}$ ($\mathcal{O}$ is closed under conjugation).

Proof. Put $z = F(1)$; applying the augmentation to $FF^* = n$ gives $z\bar{z} = n \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime, $z \in \mathfrak{p}$ or $\bar{z} \in \mathfrak{p}$. Both cannot hold: then $n = z\bar{z} \in \mathfrak{p}^2$, but $v_p(n) = v_p(n) = 1$ because p is unramified in $\mathcal{O}$. (This does not use $\mathfrak{p} \neq \overline{\mathfrak{p}}$; if $\mathfrak{p} = \overline{\mathfrak{p}}$, it shows that no such F exists. This is Turyn's self-conjugate case.)

Since $\mathfrak{p}$ lies above p , the field $\mathbb{F} = \mathcal{O}/\mathfrak{p}$ has characteristic p , so $T^p - 1 = (T - 1)^p$ in $\mathbb{F}[T]$ and $(\mathcal{O}/\mathfrak{p})[C_p] \cong \mathbb{F}[T]/(T - 1)^p$ is a local ring whose maximal ideal is the augmentation ideal $(T - 1)$; an element is a unit if and only if its augmentation is nonzero in $\mathbb{F}$. Reduce $FF^* = n \equiv 0$ modulo $\mathfrak{p}$. If $z \notin \mathfrak{p}$, the reduction of F is a unit, so $F^* \equiv 0 \pmod{\mathfrak{p}}$, i.e. $\bar{c} \in \mathfrak{p}$ for every coefficient c of F , i.e. $c \in \overline{\mathfrak{p}}$. If $z \in \mathfrak{p}$, then $\bar{z} \notin \mathfrak{p}$, the reduction of F^* is a unit, and $F \equiv 0 \pmod{\mathfrak{p}}$. $\square$

2.4. Proof of Theorem 1.1. Assume X exists, with the notation of Section 2.2. For every $\chi \in \widehat{H}$, Lemma 2.1 applies to $F = X_\chi$ and $n = |G|$. The coefficients of X_χ are the values $\chi(c_k)$, so

$$(2) \quad \chi(c_k) \overline{\chi(c_k)} \in \mathfrak{p} \quad \text{for all } \chi \in \widehat{H} \text{ and all } k \in \mathbb{Z}/p.$$

Now fix one $k \in \mathbb{Z}/p$. Parseval's identity on H gives

$$\sum_{\chi \in \widehat{H}} |\chi(c_k)|^2 = |H| \sum_{x \in H} |c_k(x)|^2 = |H|^2,$$

since every coefficient $c_k(x)$ is a root of unity. By (2) the left side lies in $\mathfrak{p}$. Hence $|H|^2 \in \mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, contradicting $p \nmid |H|$. $\square$

Example 2.2. Let $G = C_{14}$, $h = 7$ and $p = 2$. Then $H = C_7$, $L = 7$, and 2 splits in $\mathbb{Z}[\zeta_7]$ into $\mathfrak{p} \neq \bar{\mathfrak{p}}$ (the order of 2 modulo 7 is 3, and $-1 \notin \langle 2 \rangle \leq (\mathbb{Z}/7)^\times$), so Turyn's self-conjugacy argument does not apply. For a perfect μ_7 -sequence of length 14 we would have $X_\chi = a_\chi + b_\chi T \in \mathbb{Z}[\zeta_7][C_2]$ with $|a_\chi|^2 + |b_\chi|^2 = 14$ and $a_\chi \bar{b}_\chi + \bar{a}_\chi b_\chi = 0$. The column lemma says that a_χ, b_χ lie both in $\mathfrak{p}$ or both in $\bar{\mathfrak{p}}$; the Parseval step for the column c_0 , with 7 entries in μ_7 , gives $\sum_\chi |a_\chi|^2 = 49 \in \mathfrak{p} \cap \mathbb{Z} = 2\mathbb{Z}$, a contradiction. (This length is covered by the results of Ma and Ng on lengths $2p^s$ [MN09], as summarized in its abstract.)

2.5. Remarks on the proof.

Remark 2.3 (Parseval on a single coset). Summing (2) over k gives $p|H|^2 \in \mathfrak{p}$, no contradiction; Parseval on G only returns $|G|^2$. The character values of X are the $X_\chi(\theta)$, $\theta \in \mu_p$, and from them alone one gets congruences between augmentations, as in [LSZ23, Lemma 3.11]. The roots of unity enter only through $\sum_x |c_k(x)|^2 = |H|$, and the hypothesis on h only through $v_{\mathfrak{p}}(|G|) = 1$: if p ramifies in $\mathbb{Q}(\zeta_h)$, then $v_{\mathfrak{p}}(|G|) \geq 2$ and the first step of Lemma 2.1 fails. Section 4 treats a family in this regime.

Remark 2.4 (Sharpness). Neither hypothesis of Theorem 1.1 can be dropped in general.

- (a) p ramified in $\mathbb{Q}(\zeta_h)$. Perfect sequences of length p over μ_p exist for odd p [FZ62, Chu72]; $(1, i)$ is a perfect sequence of length 2 over μ_4 ; with first entry 1 there are 12 perfect μ_{12} -sequences of length 6 and 432 perfect μ_6 -sequences of length 12 [Comp].
- (b) $\nu_p(|G|) = 2$. There are 288 perfect functions $C_2^2 \times C_3 \rightarrow \mu_3$ with value 1 at 0 [Comp] ($\nu_2 = 2$, $2 \nmid 3$); Menon–Hadamard difference sets in $C_2^2 \times C_3^2$ give perfect binary arrays with $\nu_3 = 2$ [Men62, DJ96]; and Do Duc [DD19b, Thm. 4.3] constructs $\text{BH}(R \times R, h)$ for finite local rings R with residue field of order p^d whenever $p^d \in p_1\mathbb{N} + \dots + p_r\mathbb{N}$, where $p_1, \dots, p_r$ are the prime divisors of h ; for $R = \mathbb{F}_5$ and $h = 6$ this gives a $\text{BH}(C_5 \times C_5, 6)$ with $5 \nmid 6$. In all these examples the Sylow subgroup for the prime in question is *not cyclic*. Whether the conclusion persists for $\nu_p(|G|) = 2$ when the Sylow p -subgroup is C_{p^2} is open.

As a check (not used in the proof), exhaustive searches find no perfect function in the non-cyclic cases $C_2 \times C_6$ ($h = 4, 8$), $C_3 \times C_6$ ($h = 3, 6$; for $h = 6$, $p = 2$ and $h \equiv 2 \pmod{4}$), $C_2 \times C_{10}$ ($h = 4, 6$), $C_2^3 \times C_3$ ($h = 4$), in the cases $p = 2$, h odd, C_6 ($h = 3$) and C_{10} ($h = 5$), and in C_{14} ($h = 14$), C_{15} ($h = 5$), C_{18} ($h = 6$), while reproducing the positive counts above [Comp].

Remark 2.5 (The case $H = 1$). For $G = C_p$ the proof reduces to: X or X^* vanishes modulo $\mathfrak{p}$, so a root of unity lies in $\mathfrak{p}$. This case is [DDS19, Thm. 3.5] for $a = 1$, proved there with the description of cyclotomic integers of norm p^b . It is also implied by the theorem of Leung, Chue and Zhao [LCZ25] that a cyclic Butson matrix of prime order is equivalent to a Fourier matrix (Section 5).

2.6. Formal statement. Theorem 1.1 is formalized in Lean 4 with Mathlib (Section 6.1). The general statement reads, verbatim (namespace `OAI.CirculantHadamard.TheoremS`):

```
theorem no_perfect_function_sharp (G : Type*) [AddCommGroup G] [Fintype G] (h p : ℕ)
  (hp : p.Prime) (hpG : p | Fintype.card G) (hp2 : ¬ p ^ 2 | Fintype.card G)
  (hunr : ¬ p | h ∨ (p = 2 ∧ ¬ 4 | h)) (a : G → ℂ) (ha : ∀ g, a g ^ h = 1)
  (hperf : ∀ s ≠ 0, ∑ g, a (g + s) * conj (a g) = 0) : False
```

The value $h = 0$ is excluded by `hunr`. Special cases are stated separately (Table 1). The formal proof follows the proof above; in particular it does not use $\mathfrak{p} \neq \bar{\mathfrak{p}}$.

3. CONSEQUENCES

3.1. Relative difference sets. A k -subset D of a group G of order mn is an (m, n, k, λ) -relative difference set (RDS) relative to a subgroup N of order n if $DD^{(-1)} = k + \lambda(G - N)$ in $\mathbb{Z}[G]$ [Pot95]; it is *semi-regular* if $k = m$.

Corollary 3.1. *Let G be a finite abelian group, $N \leq G$ a subgroup with $|N| = n$ and $|G/N| = m$, and let D be a semi-regular $(m, n, m, m/n)$ -RDS in G relative to N . If a prime p satisfies $p \parallel m$, then N is a p -group. [Lean]*

Proof. Suppose N is not a p -group. A semi-regular RDS meets every coset of N exactly once; let $\sigma : G/N \rightarrow G$ be the section with image D , and $\lambda = m/n$. Choose $x \in N$, $x \neq 0$, with $p \nmid \text{ord } x$, and $\psi \in \widehat{G}$ with $\psi(x) \neq 1$. Let $p^v \parallel |G|$ and $\varphi = \psi^{p^v}$. Then $\varphi(x) \neq 1$, since p^v is prime to the order of $\psi(x)$, and the order of φ divides $h := |G|/p^v$, which is prime to p . Put $a(q) = \varphi(\sigma(q)) \in \mu_h$ for $q \in G/N$. For $s \neq 0$ in G/N , every g in the coset $\tilde{s} + N$ of s is a difference $\sigma(q + s) - \sigma(q)$ for exactly λ values of q , and no other element is, so

$$\sum_{q \in G/N} a(q + s) \overline{a(q)} = \lambda \sum_{g \in \tilde{s} + N} \varphi(g) = \lambda \varphi(\tilde{s}) \sum_{u \in N} \varphi(u) = 0,$$

because φ is a nontrivial character of N . Thus a is a perfect function $G/N \rightarrow \mu_h$ with $p \parallel |G/N|$ and $p \nmid h$, which Theorem 1.1 excludes. $\square$

Example 3.2. Let p be an odd prime, $G = C_p \times C_4^2$ and $N = 2C_4^2 \cong C_2^2$, so $n = 4$ and $m = 4p$. Here N is not a direct factor of G and $\exp(G) = 4p$ divides m , so neither a splitting hypothesis nor the exponent bound [Pot95, Thm. 4.1.1] applies; Corollary 3.1 (with the prime p) excludes a semi-regular $(4p, 4, 4p, p)$ -RDS in G relative to N . For $p = 5$ an exhaustive search finds none [Comp].

The conclusion cannot be strengthened to “no such RDS”: in $C_2^2 \times C_3^2$, relative to a subgroup of order 3, there are semi-regular $(12, 3, 12, 4)$ -RDS, the case $p = 2$, $t = 1$ of the family of Davis, Jedwab and Mowbray [DJM98] (as recalled in [LSZ23, Sect. 1]); here $3 \parallel 12$ and N is a 3-group. Comparison with [LSZ23]. In the notation of [LSZ23], an (mn, n, mn, m) -RDS has a forbidden subgroup of order n and $|G/N| = mn$. If $p \nmid n$ and $n \geq 2$, then $p \parallel pn = |G/N|$ while N is not a p -group, so Corollary 3.1 shows that no abelian (pn, n, pn, p) -RDS exists. Hence Corollary 3.1 contains every nonexistence statement of [LSZ23, Sect. 6], which is about such RDS: Theorem 6.1 (for all distinct primes p, q and $r \geq 1$, without any of its conditions (a)–(c)); Theorem 6.2, whose hypotheses ($n > 1$ prime to p) already exclude the RDS; Corollary 6.3, in the stronger form that an abelian (pn, n, pn, p) -RDS with $n \geq 2$ can exist only if $p \mid n$; Corollary 6.4, in the range $\gcd(p, n) = 1$, $n \geq 2$, in which it is derived there from Theorem 6.2;² and Theorem 6.5, without the assumptions that p, q are odd and $\gcd(p, q - 1) = 1$. This includes the theorem of Ma [Ma96] on (pq, q, pq, p) -RDS with $p > q$, as stated in [LSZ23, Sect. 1]. For RDS whose index is divisible by p exactly once and whose forbidden subgroup is divisible by another prime, Corollary 3.1 is the case $b = 1$ of [LSZ23, Thm. 5.1(a)] without its self-conjugacy condition. It decides the following entries marked “?” in the tables of [LSZ23]: Table 1, $(2n, n, 2n, 2)$, $n = 24, 96$ (prime 3); Table 2, $(3n, n, 3n, 3)$, $n = 13$ (prime 3) and $n = 39$ (prime 13); Table 3, $(4n, n, 4n, 4)$, $n = 12, 20, 48, 80$ (primes 3, 5, 3, 5). The other five entries marked “?” have no prime dividing $|G/N|$ exactly once.

3.2. The conjectures of Mow and of Do Duc.

Corollary 3.3. *Let G be a finite abelian group of order n and suppose that a $\text{BH}(G, h)$ exists. Then (i) $p \mid h$ for every prime p with $\nu_p(n) = 1$, and (ii) $\nu_2(h) \geq 2$ if $n \equiv 2 \pmod{4}$. [Lean]*

Proof. (i) is Theorem 1.1 with $p \nmid h$. (ii) If $n \equiv 2 \pmod{4}$, then $2 \parallel n$; if $\nu_2(h) \leq 1$, then 2 is unramified in $\mathbb{Q}(\zeta_h)$ and Theorem 1.1 applies with $p = 2$. $\square$

²Read literally, the hypothesis of [LSZ23, Cor. 6.4] also admits $n = 1$ and some n divisible by p , where it is not covered by its proof.

In Mow's formulation this is the necessity of the factor p of $\sqrt{rn}$ for every prime p with $\nu_p(n) = 1$ and of the factor 2 for $n \equiv 2 \pmod{4}$. Corollary 1.2 follows, since for square-free n conditions (i), (ii) say $n \mid h$ and, for even n , $4 \mid h$, and these suffice by [DDS19, Cor. 2.5] (see also [DD19a, Result 4.1]). Before, the necessity of (ii) was known for $n = 2$ [DDS19, Thm. 3.6], $h = 2$, $n = 2p^c$ with $h = 2p^b$ [DD19a, Cor. 3.12], and when 2 is self-conjugate modulo the odd part of h (Turyn's argument [Tur65]; [KSW85, Prop. 6]).

Do Duc's list. Of the 2687 pairs (n, h) , $n, h \leq 100$, listed as open in [DD19a, Remark 3.13], Theorem 1.1 excludes 2323: 2262 with a prime p , $\nu_p(n) = 1$, $p \nmid h$, and 61 further pairs with $n \equiv h \equiv 2 \pmod{4}$ [Comp], leaving 364 pairs such as $(18, 8)$ and $(50, 4)$. The list is not closed under earlier results, so we checked the 2323 pairs against the earlier criteria we could identify: Turyn's self-conjugacy argument applied to the principal character (if a prime p with $\nu_p(n)$ odd is unramified in $\mathbb{Q}(\zeta_h)$ and self-conjugate modulo h , then $|\chi_0(X)|^2 = n$ is impossible in $\mathbb{Z}[\zeta_h]$); the case $h = 4$, by Arasu's survey [Ara11, Remark after Thm. 4.11]; prime h , by Yayla's table [Yay16, Table 2] together with [FX08] and [LSZ23, Thm. 6.5]; the computations of Hiranandani and Schlenker for $n, h \leq 15$ [HS16, Table 1]; and the results on generalized bent functions of type $[1, q]$ recalled in Section 3.3. They cover 237 of the 2262 pairs (e.g. $(10, 4)$ [Tur70] and $(24, 4)$) and 23 of the 61 pairs. We also applied Schmidt's field-descent bound [Sch99, LS05] to the remaining pairs, in the form (C4) of Appendix A (a character of order d for every $d \mid n$, $L = \text{lcm}(h, d)$, $C = nh/L$), with both descent exponents [LS05, Def. 2.6] and [LS12, Def. 2.10]: it excludes none of them [Comp]. The remaining $2025 + 38 = 2063$ pairs are not excluded by any result known to us, subject to [LCZ25] (Section 5): $463 + 11 = 474$ of them have $n = 2p$ with p an odd prime, 17 of these with $n = 6$. Examples with $n \neq 2p$ are $(15, 12)$, $(21, 9)$, $(39, 13)$, $(30, 16)$ and $(30, 90)$. Our list of earlier criteria is not claimed to be complete.

3.3. Generalized bent functions of type $[1, q]$. A generalized bent function of type $[m, q]$ is a function $\mathbb{Z}_q^m \rightarrow \mathbb{Z}_q$ with all Fourier coefficients of absolute value $q^{m/2}$, i.e. a BH($\mathbb{Z}_q^m, q$). Kumar, Scholtz and Welch [KSW85] constructed them unless m is odd and $q \equiv 2 \pmod{4}$. For $m = 1$, Theorem 1.1 applies with $p = 2$:

Corollary 3.4. *For every $q \equiv 2 \pmod{4}$ there is no generalized bent function of type $[1, q]$, i.e. no perfect sequence of length q over μ_q . [Lean-derived]*

In Lean only the general statement `no_perfect_sequence_two_mod_four` is verified; the translation to generalized bent functions is not formalized.

As stated in [YD26, Sect. 1], [LYZ26, Sect. 1] and [LZ18, Sect. 1] (we have not seen the full texts of [Pei93, Ike99, Fen01, LMF02, FL03b]), the case $m = 1$, $q = 2N$, was treated for 2 self-conjugate modulo N [KSW85, Prop. 6], $N = 7$ [Pei93], every prime divisor of N self-conjugate modulo N [Ike99], $N = p^l$ and $N = p_1^a p_2^b$ under congruence conditions [Fen01, LMF02, FL03a, FL03b], $N = p^a$ [LS19], and $N = 3^a 7^b$ [YD26, Thm. 5]; Corollary 3.4 contains these statements as stated there. The closest argument we found is [LYZ26, Prop. 1.3], which uses that 2 is unramified in $\mathbb{Q}(\zeta_N)$ to show that certain Fourier coefficients are not roots of unity. For odd $m \geq 3$, $2^m \mid q^m$ and Theorem 1.1 says nothing; [LYZ26] constructs functions of type $[m, q]$, e.g. [3, 14].

3.4. Circulant complex Hadamard matrices.

Corollary 3.5 (Arasu's list). *There is no circulant BH($N, 4$) for $N \in \{260, 340, 442, 468, 520, 580, 680, 754, 820, 884, 890\}$. [Lean-derived] Consequently, by [Ara11, Remark after Thm. 4.11], a circulant complex Hadamard matrix of order $N \leq 1000$ exists only for $N \in \{1, 2, 4, 8, 16\}$.*

Proof. $260 = 2^2 \cdot 5 \cdot 13$, $340 = 2^2 \cdot 5 \cdot 17$, $442 = 2 \cdot 13 \cdot 17$, $468 = 2^2 \cdot 3^2 \cdot 13$, $520 = 2^3 \cdot 5 \cdot 13$, $580 = 2^2 \cdot 5 \cdot 29$, $680 = 2^3 \cdot 5 \cdot 17$, $754 = 2 \cdot 13 \cdot 29$, $820 = 2^2 \cdot 5 \cdot 41$, $884 = 2^2 \cdot 13 \cdot 17$, $890 = 2 \cdot 5 \cdot 89$:

each has an odd prime divisor of exponent 1. Apply Theorem 1.1 with $h = 4$ (in Lean only the general statement `no_perfect_quaternary_sequence` is verified, not the eleven instances). The second sentence rests on Arasu's remark that these eleven orders are the only ones ≤ 1000 not excluded before, and on the existence of the matrices of orders 1, 2, 4, 8, 16. $\square$

3.5. Equality in the descent bound. For larger orders we combine Theorem 1.1 with the classical criteria and the following refinement of the equality case of Schmidt's field-descent bound.

Lemma 3.6. *Let $F \geq 2$, $C > 0$, and $Y = \sum_{i=0}^{F-1} d_i \zeta_F^i$ with integers $0 \leq d_i \leq C$ and $Y\bar{Y} = n \in \mathbb{Z}$. Then $n \leq C^2 F^2 / (4\varphi(F))$, and the inequality is strict if F has an odd prime divisor.*

Proof. Let $D(x) = \sum_i d_i x^i$ and $D'(x) = D(x) - \frac{C}{2} \sum_i x^i$. For t prime to F , $D'(\zeta_F^t) = D(\zeta_F^t)$ has absolute value $\sqrt{n}$, because n is rational and complex conjugation commutes with $\text{Gal}(\mathbb{Q}(\zeta_F)/\mathbb{Q})$. Hence

$$\varphi(F) n \leq \sum_{t \in \mathbb{Z}/F} |D'(\zeta_F^t)|^2 = F \sum_i \left(d_i - \frac{C}{2}\right)^2 \leq F^2 \frac{C^2}{4}.$$

Equality forces (a) $d_i \in \{0, C\}$ for all i , so $D = C E$ with $E = \sum_{i \in S} x^i$ for some $S \subseteq \mathbb{Z}/F$, and (b) $D'(\zeta_F^t) = 0$ for every t not prime to F . Let r be an odd prime divisor of F . By (b), $E - \frac{1}{2}J$ vanishes at every character of C_F whose order divides F/r , i.e. at every character of C_F/C_r ; by Fourier inversion the image of E in $\mathbb{Q}[C_F/C_r]$ equals $\frac{r}{2}J_{C_F/C_r}$. But this image has integer coefficients and $r/2 \notin \mathbb{Z}$. $\square$

The inequality is the argument behind Schmidt's bound [Sch99, LS05]; we have not checked whether the equality analysis appears in [Sch02] or later work. It applies to circulant $\text{BH}(N, 4)$, $4 \mid N$, as follows. Let a be a perfect quaternary sequence of length N , χ a character of C_N of order N , and $X = \chi(X_a) = \sum_{t \in \mathbb{Z}/N} c_t \zeta_N^t$, where c_t counts the g with $a_g \chi(g) = \zeta_N^t$; then $0 \leq c_t \leq C := 4$. By field descent [Sch99, LS05] (see [DD19a, Result 2.8]) there is j with $X \zeta_N^j \in \mathbb{Z}[\zeta_F]$, where $F = F(N, N)$ divides N and is divisible by every prime divisor of N . Since $1, \zeta_N, \dots, \zeta_N^{N/F-1}$ is a basis of $\mathbb{Q}(\zeta_N)/\mathbb{Q}(\zeta_F)$, $X \zeta_N^j = \sum_{i < F} d_i \zeta_F^i$ with $d_i = c_{iN/F-j}$. Lemma 3.6 gives Schmidt's bound $N \leq 4F^2/\varphi(F)$ and excludes equality when F has an odd prime divisor.³

For $N = 1800 = 2^3 \cdot 3^2 \cdot 5^2$: $m_2 = 15$, $m_3 = 20$, $m_5 = 12$, $\text{ord}_{15}(2) = \text{ord}_{20}(3) = 4$, $\text{ord}_{12}(5) = 2$, so $b_2 = 3$, $b_3 = b_5 = 1$, $F = 120$ and $4F^2/\varphi(F) = 1800$: equality, with $3, 5 \mid F$. In this way Lemma 3.6 excludes, among the lengths $N \equiv 0 \pmod{4}$, $1000 < N \leq 10^6$, that pass all other criteria of Corollary A.1, exactly the seven lengths 1800, 3600, 6084, 12100, 12168, 14112, 24200 (Table 2) [Comp]; Theorem 1.1 does not apply to them.

3.6. Further special cases. Of the eleven pairs (n, p) , $n \leq 100$, left undecided in Yayla's table [Yay16] of perfect p -ary sequences, Theorem 1.1 excludes $(39, 13)$, $(55, 11)$, $(63, 3)$, $(84, 3)$ (primes 3, 5, 7, 7) and $(33, 11)$, $(69, 23)$, $(95, 19)$, already excluded by [LSZ23, Thm. 6.5]; $(28, 7)$ and $(92, 23)$ were excluded in [FX08], and $(56, 7)$, $(99, 11)$ are excluded by Theorem 4.1. The pair $(63, 3)$ does not occur in Do Duc's list [DD19a, Remark 3.13], although none of the results from which that list is derived ([DD19a, Results 1.1–1.4, Thms. 3.2, 3.3, 3.7, 3.11]) excludes it as stated [Comp]; the same holds for $(42, 3)$, $(78, 3)$ (lengths $3pq$ with p or $q \leq 3$) and $(14, 33)$, $(20, 51)$, $(26, 69)$ (pairs $(3+q, 3q)$), which suggests that [DD19a, Results 1.1(2),(3)] were applied

³Here $F(m, n)$ is the descent exponent of [LS05, Def. 2.6], equivalently [LS12, Def. 2.10]: for $m = \prod_i p_i^{c_i}$ and a prime $q \mid n$ let $m_q = \prod_{p_i \neq q} p_i$ if m is odd or $q = 2$, and $m_q = 4 \prod_{p_i \neq 2, q} p_i$ otherwise; then $F(m, n) = \prod_i p_i^{b_i}$ with each $b_i \geq 1$ minimal such that for every prime $q \mid n$ one of the following holds: $q = p_i$ and $(p_i, b_i) \neq (2, 1)$; $b_i = c_i$; or $q \neq p_i$ and $q^{\text{ord}_{m_q}(q)} \not\equiv 1 \pmod{p_i^{b_i+1}}$. The original [Sch99, Def. 3.1] takes $m_q = 4 \prod_{p_i \neq 2, q} p_i$ for every even m , including $q = 2$, where $\text{ord}_{m_q}(q)$ is not defined; we use the form of [LS05, LS12].

without the restriction $p, q > 3$. The pair $(63, 3)$ is listed as undecided in Yayla's table [Yay16]; we do not know whether it was excluded earlier. Our counts refer to the published list and are not affected. The following are special cases of Theorem 1.1. (1) Lv [Lv17, Thm. 1.3]: no perfect p -ary sequence of length $p^a q n'$ for $p \equiv 5 \pmod{8}$, $p > p_0$, primes q with $\text{ord}_p(q) = (p-1)/4$, and $n' = 1$ or $(\frac{p'}{p}) = -1$ for all primes $p' \mid n'$; here q is a quadratic residue modulo p , so $q \parallel p^a q n'$ (prime q). (2) Ma and Ng [MN09], as summarized in its abstract: no perfect p -ary sequence of length pq , $q > p$ (prime q), or $2p^s$ (prime 2). (3) Arasu, de Launey and Ma, as stated in [Ara11, Thm. 4.7]: a circulant complex Hadamard matrix of order $2tp$, $p \equiv 1 \pmod{4}$, t odd, $(t, p) = 1$, forces $t \geq p/(\alpha + 1)$ for a certain α ; Theorem 1.1 (prime p) shows that it does not exist.

4. THE RAMIFIED REGIME

Theorem 1.1 needs a prime p with $p \parallel |G|$ that is unramified in $\mathbb{Q}(\zeta_h)$. We treat the simplest family to which it does not apply, the perfect p -ary sequences of length pq^e .

Theorem 4.1. *Let $p \neq q$ be primes and $e \geq 1$.*

- (i) *There is no perfect sequence of length pq^e over μ_p , i.e. no circulant $\text{BH}(pq^e, p)$. [Lean]*
- (ii) *More generally, let $u \geq 1$ with $\gcd(u, pq) = 1$ such that every prime divisor of u is self-conjugate modulo pq (Section 4.1). Then there is no perfect sequence of length $pq^e u$ over μ_p .*

What is new. For $e = 1$, Theorem 4.1(i) follows from Theorem 1.1 with the prime q , and for $p = 2$ from Theorem 1.1 with the prime 2. If q is self-conjugate modulo p , then q is self-conjugate modulo $\text{lcm}(pq^e, p)$, and [DD19a, Thm. 3.3] excludes $\text{BH}(pq^e, p)$ for every e ; e.g. $(75, 3)$, where $5 \equiv -1 \pmod{3}$. So the new range of (i) is: p odd, $e \geq 2$ and q not self-conjugate modulo p . There the only prime dividing pq^e exactly once is p , which ramifies in $\mathbb{Q}(\zeta_p)$, so the methods based on a prime dividing the length exactly once do not apply, and the self-conjugacy argument at q ([DD19a, Thm. 3.3]) does not apply. Some of these lengths are covered by other earlier criteria (Section 4.5): of the 334 lengths $pq^e \leq 2000$ with p odd and $e \geq 2$, 300 are excluded by them and 34 are not [Comp], namely 56, 99, 112, 117, 184, 224, 248, 275, 297, 351, 448, 475, 496, 584, 621, 736, 775, 847, 891, 896, 931, 992, 1053, 1269, 1375, 1421, 1472, 1504, 1593, 1775, 1792, 1813, 1863, 1984. In (ii), [DD19a, Thm. 3.3] applies exactly when q or some prime divisor r of u is self-conjugate modulo $\text{lcm}(pq^e u, p) = pq^e u$ in the sense of [DD19a, Def. 2.5], i.e. when q is self-conjugate modulo pu or r is self-conjugate modulo $pq^e u_{r'}$, $u_{r'}$ the largest divisor of u prime to r ; the latter is the case if q is odd and u is a prime power, since then $r^j \equiv -1 \pmod{pq}$ implies $r^{jq^{e-1}} \equiv -1 \pmod{pq^e}$. An instance of (ii) not covered by the criteria of Section 4.5 is $(1400, 7)$, $1400 = 7 \cdot 2^3 \cdot 5^2$: 5 is self-conjugate modulo 14 but not modulo 56, and 2 is not self-conjugate modulo 175.

Corollary 4.2. *There is no perfect 7-ary sequence of length 56, 847 or 1400, and no perfect 11-ary sequence of length 99.*

The pairs $(56, 7)$ and $(99, 11)$ are the two pairs of Yayla's table [Yay16] not covered by Theorem 1.1 or earlier work (Section 3.6).

The main new ingredient is a comparison lemma at the ramified prime p (Lemma 4.10): for $F \in \mathcal{O}[C_p]$ with $FF^* = n$, $v_p(n) = 1$, over $\mathcal{O} = \mathbb{Z}[\zeta_{pd}]$, the valuation of $F(\zeta_p^j)$ at a prime above p does not depend on j , and $F(\zeta_p^j) \equiv F(1)$ to the next order. With it at p and a classical comparison at q , an alternating product of character values is 1 (Theorem 4.12); this forces proportional projections and a contradiction modulo $1 - \zeta_p$.

4.1. Preliminaries. We use the notation of Section 2.1. For a quotient map $V \rightarrow V/U$ we write ϖ_U (or ϖ_t when the image is cyclic of order t) for the induced map of group rings.

Proposition 4.3. *Let $G = N \times H$ be abelian with $N \cong C_r$, r prime. The semi-regular $(|H|, r, |H|, |H|/r)$ -RDS in G relative to N are the graphs $\{(f(x), x) : x \in H\}$ of the maps $f : H \rightarrow N$ such that $\psi \circ f$ is perfect for one (equivalently every) character $\psi \neq 1$ of N .*

Proof. A semi-regular RDS meets each coset of N once, so it is a graph. For a character $\psi \otimes \chi$ of $N \times H$ with $\psi \neq 1$ the character value of the graph is $\chi(X_{\psi \circ f})$, and $|\cdot|^2 = |H|$ for all such characters is the RDS condition. The maps $\psi \circ f$, $\psi \neq 1$, are Galois conjugate. $\square$

Proposition 4.4 ([Pot95, Thm. 4.1.1]). *If R is an abelian $(m, n, m, m/n)$ -RDS in G , then $\exp(G) \mid m$, or $G = C_4$ and $n = 2$.*

A prime r is *self-conjugate modulo L* if $r^j \equiv -1 \pmod{L_{r'}}$ for some $j \geq 0$, where $L_{r'}$ is the largest divisor of L prime to r .

Lemma 4.5 ([Tur65]). *If r is self-conjugate modulo L , every prime $\mathfrak{Q}$ of $\mathbb{Z}[\zeta_L]$ above r satisfies $\bar{\mathfrak{Q}} = \mathfrak{Q}$; hence $\alpha\bar{\alpha} = n \in \mathbb{Z}_{>0}$ implies $v_{\mathfrak{Q}}(\alpha) = \frac{1}{2}v_{\mathfrak{Q}}(n)$.*

Proof. The decomposition group of r in $(\mathbb{Z}/L)^\times$ is generated by the inertia group and r [Was97, Thm. 2.13]; it contains -1 iff $-1 \in \langle r \rangle$ modulo $L_{r'}$. $\square$

Lemma 4.6. *Let q be a prime, η a root of unity of order prime to q , and A the localization of $\mathbb{Z}[\eta]$ at a maximal ideal containing q . Then A is a discrete valuation ring with maximal ideal qA . If ξ is a primitive q^e -th root of unity and $D = (q-1)q^{e-1}$, then $A[\xi]$ is a discrete valuation ring, free over A with basis $1, \xi, \dots, \xi^{D-1}$, with maximal ideal $(\xi-1)$, and $\sum_{j < D} a_j \xi^j \in q^s A[\xi]$ iff all $a_j \in q^s A$.*

Proof. q is unramified in $\mathbb{Z}[\eta]$, so A is a DVR with uniformizer q . $\Phi_{q^e}(1+Y)$ is Eisenstein over A , so $A[\xi] \cong A[X]/(\Phi_{q^e})$ is free with the stated basis, and $A[\xi]/qA[\xi] \cong (A/qA)[Y]/(Y^D)$ is local with principal maximal ideal (Y) . Since $A[\xi]$ is finite over A , every maximal ideal of $A[\xi]$ contains q ; hence $A[\xi]$ is local with maximal ideal $(\xi-1)$, which contains q , and is a DVR. $\square$

Lemma 4.7 (Unramified comparison). *Let A be as in Lemma 4.6, ρ a primitive q -th root of unity, $e \geq 1$, $0 \leq f \leq e$, and $F, K' \in A[C_{q^e}]$ with $FK' = q^f b$, $b \in A^\times$. Then $F(\rho)/F(1)$ and $K'(\rho)/K'(1)$ are units of $A[\rho]$ with residue 1.*

Proof. All character values divide $q^f b$ and are nonzero, and $K'(\rho)/K'(1) = F(1)/F(\rho)$, so it suffices to treat F or K' . If $f \leq 1$, then $v_q(F(1)) + v_q(K'(1)) = f \leq 1$, so one of $F(1)$, $K'(1)$ is a unit of A , say $F(1)$; since $F(\rho) \equiv F(1) \pmod{\rho-1}$ and $A[\rho]$ is local with maximal ideal $(\rho-1)$, $F(\rho)$ is a unit and $F(\rho)/F(1) \equiv 1$. If $f \geq 2$, then $e \geq 2$. For a primitive q^e -th root ξ , $v(F(\xi)) + v(K'(\xi)) = fD \geq 2D$ in the DVR $A[\xi]$ (with $v(\xi-1) = 1$, $v(q) = D$), so after interchanging F and K' we have $F(\xi) \in qA[\xi]$. By Lemma 4.6 the remainder of F modulo $\Phi_{q^e}(T)$ has coefficients in qA , i.e. $F \equiv \Phi_{q^e}(T)S \pmod{q}$ for some S . Let $\varpi : A[C_{q^e}] \rightarrow A[C_{q^{e-1}}]$ be the projection; then $\varpi(\Phi_{q^e}(T)) = q$, so $\varpi(F) \in qA[C_{q^{e-1}}]$. The pair $(\varpi(F)/q, \varpi(K'))$ satisfies $(\varpi(F)/q)\varpi(K') = q^{f-1}b$ with $f-1 \leq e-1$, and its values at 1 and ρ are those of $(F/q, K')$, since ρ factors through $C_{q^{e-1}}$. We conclude by induction on f . $\square$

The case $f \leq 1$ is the first-order congruence also used in [LSZ23, Lemma 3.11]. The descent for $f \geq 2$ is Ma's lemma [Ma85], [MS95, Lemma 3.1], [DD19a, Result 2.4] for a single character: $\Phi_{q^e}(T) = \sum_{i < q} T^{iq^{e-1}}$ is the sum of the elements of the subgroup P of order q of $C_{q^e} = \langle T \rangle$, so $F \equiv \Phi_{q^e}(T)S \pmod{q}$ is the conclusion $F = qX + PY$ of that lemma; see also [Ope26, Lemma 3.1]. Lemma 4.7 is not formalized as stated (Section 6.1).

Lemma 4.8. (i) *An element of $\mathbb{Z}[\zeta_M]$ of absolute value 1 under every embedding is a root of unity [Kro57].* (ii) *In a local domain of residue characteristic ℓ , a root of unity with residue 1 has ℓ -power order.*

Proof. (i) is Kronecker's theorem. (ii) If $\zeta \neq 1$ has order s prime to ℓ and residue 1, then $0 = 1 + \zeta + \cdots + \zeta^{s-1}$ reduces to $s \neq 0$ in the residue field. In general, if ζ has order $\ell^a s$ with $\ell \nmid s$, then ζ^{ℓ^a} has order s and residue 1, so $s = 1$ by the case just treated. $\square$

4.2. The ramified comparison lemma. In $\text{BH}(pq^e, p)$ with p odd and $e \geq 2$, the prime q divides the order $e \geq 2$ times, so Lemma 2.1 is not available at q , and at p the coefficient ring $\mathbb{Z}[\zeta_p]$ is ramified.

Example 4.9. Lemma 4.7 with $f \leq 1$, and [LSZ23, Lemma 3.11], use that one of $F(1)$, $F^*(1)$ is a unit at the prime in question, i.e. $v_{\mathfrak{P}}(F(1)) \in \{0, v_{\mathfrak{P}}(n)\}$. This fails at a ramified prime, already on existing objects. Circulant $\text{BH}(20, 10)$ exist (400 perfect μ_{10} -sequences of length 20 with first entry 1). Write such a sequence on $C_5 \times C_4$ and let $F \in \mathbb{Z}[\zeta_{20}][C_5]$ be its partial evaluation at a character of C_4 ; then $FF^* = 20$ and $v_5(20) = 1$, but at both primes $\mathfrak{P}$ of $\mathbb{Q}(\zeta_{20})$ above 5 (ramification index 4) one has $v_{\mathfrak{P}}(F(1)) = 2$, for all 400 sequences and all four characters [Comp]. For general $F \in \mathbb{Z}[\zeta_{40}][C_5]$ with $FF^* \in \{5, 10\}$ the pairs $(v_{\mathfrak{P}}(F(1)), v_{\overline{\mathfrak{P}}}(F(1)))$ take the values $(1, 3)$, $(2, 2)$, $(3, 1)$. Feng's argument [Fen09, Lemmas 2, 3] uses that $\mathbb{F}_p[E]$ is semisimple; the analogous ring $\mathcal{O}/p\mathcal{O}$ is not reduced when p ramifies in $\mathcal{O}$.

Let p be a prime, $d \geq 1$ with $p \nmid d$, $\mathcal{O} = \mathbb{Z}[\zeta_{pd}]$, $\zeta = \zeta_p$ and $y = \zeta - 1$. For $F = \sum_{a \in \mathbb{Z}/p} u_a T^a \in \mathcal{O}[C_p]$ and $j \in \mathbb{Z}/p$ write $F(\zeta^j) = \sum_a u_a \zeta^{ja}$; the coefficients u_a may involve ζ_p .

Lemma 4.10 (Lemma A). *Let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above p and let $F, V \in \mathcal{O}[C_p]$ satisfy $FV = n$ with $n \in \mathcal{O}$, $v_{\mathfrak{P}}(n) = p - 1$ (for instance $n \in \mathbb{Z}$ with $v_p(n) = 1$). Then there is $\alpha \geq 0$ such that for all $j \in \mathbb{Z}/p$, including $j = 0$,*

$$v_{\mathfrak{P}}(F(\zeta^j)) = \alpha \quad \text{and} \quad F(\zeta^j) \equiv F(1) \pmod{\mathfrak{P}^{\alpha+1}};$$

in particular $F(\zeta^j)/F(1)$ is a $\mathfrak{P}$ -unit with residue 1. [Lean, $n \in \mathbb{Z}$]

No hypothesis $\mathfrak{P} = \overline{\mathfrak{P}}$ or $\mathfrak{P} \neq \overline{\mathfrak{P}}$ is needed, and V is arbitrary; for $V = F^*$ the hypothesis is $FF^* = n$.

Proof. Work in the completion $\mathcal{O}_{\mathfrak{P}} = W[\zeta]$, where W is unramified over $\mathbb{Z}_p$ with residue field $\mathbb{k}$; y is a uniformizer, normalized by $v(y) = 1$, so $v(p) = p - 1$. Since $\Phi_p(1 + y) \equiv y^{p-1} \pmod{p}$, $\mathcal{O}_{\mathfrak{P}}/p\mathcal{O}_{\mathfrak{P}} \cong \mathbb{k}[y]/(y^{p-1})$, and every $u \in \mathcal{O}_{\mathfrak{P}}$ has a unique expansion $u \equiv \sum_{k=0}^{p-2} \bar{u}_k y^k \pmod{p}$ with *digits* $\bar{u}_k \in \mathbb{k}$.

Digit polynomials. For an integer N and $0 \leq i \leq p - 2$, the binomial coefficient $\binom{N}{i}$ modulo p depends only on $N \bmod p$ and is a polynomial in N of degree i over $\mathbb{F}_p$. Hence, with $u_a \equiv \sum_k \bar{u}_{a,k} y^k$,

$$F(\zeta^j) = \sum_a u_a (1 + y)^{ja} \equiv \sum_{t=0}^{p-2} c_t(j) y^t \pmod{p}, \quad c_t(j) = \sum_a \sum_{k+i=t} \bar{u}_{a,k} \binom{ja}{i},$$

where $c_t \in \mathbb{k}[T]$ has degree $\leq t$. In the same way $V(\zeta^j) \equiv \sum_t d_t(j) y^t$ with $\deg d_t \leq t$.

Valuations. Put $a(j) = v(F(\zeta^j))$ and $b(j) = v(V(\zeta^j))$. Evaluating $FV = n$ at ζ^j gives $a(j) + b(j) = p - 1$ for all p values of j . Let $\alpha = \min_j a(j)$, $\beta = \min_j b(j)$, so $\alpha + \beta \leq p - 1$.

If $\alpha = 0$, some $F(\zeta^{j_0})$ is a unit; since $F(\zeta^j) \equiv F(1) \pmod{y}$, all $F(\zeta^j)$ are units congruent to $F(1)$ modulo y , which is the claim. If $\beta = 0$, the same holds for V , so $a(j) = p - 1$ for all j and $F(\zeta^j)/F(1) = V(1)/V(\zeta^j) \equiv 1$. So let $1 \leq \alpha, \beta \leq p - 2$.

Since $a(j) \geq \alpha$ for all j , the polynomials c_t , $t < \alpha$, vanish at all p points of $\mathbb{F}_p$; having degree $< p$, they are zero. The polynomial c_α is nonzero (it does not vanish where $a(j) = \alpha \leq p - 2$) and has degree $\leq \alpha$, so $a(j) > \alpha$ for at most α values of j ; likewise $b(j) > \beta$ for at most β values. As $\alpha + \beta < p$, some j_0 has $a(j_0) = \alpha$ and $b(j_0) = \beta$, whence $\alpha + \beta = p - 1$; then $a(j) \geq \alpha$, $b(j) \geq \beta$ and $a(j) + b(j) = p - 1$ force $a \equiv \alpha$, $b \equiv \beta$.

Leading digits. As $\alpha, \beta \leq p - 2$, the residue of $F(\zeta^j)/y^\alpha$ is $c_\alpha(j)$ (the error lies in $p\mathcal{O}_{\mathfrak{P}}$, of valuation $p - 1 > \alpha$) and that of $V(\zeta^j)/y^\beta$ is $d_\beta(j)$. Their product is the residue $\kappa \neq 0$ of n/y^{p-1} , independent of j . So $c_\alpha d_\beta - \kappa$, of degree $\leq \alpha + \beta = p - 1$, vanishes at all p points of $\mathbb{F}_p$ and is zero; since $\mathbb{k}[T]$ is a domain, c_α is a constant. Hence $F(\zeta^j)/y^\alpha$ and $F(1)/y^\alpha$ have the same nonzero residue, i.e. $F(\zeta^j) \equiv F(1) \pmod{\mathfrak{P}^{\alpha+1}}$. $\square$

Remark 4.11. (a) *All p points are needed.* The degree bound $p - 1$ is beaten only because j runs over all of $\mathbb{F}_p$, including $j = 0$, and the comparison is with $F(1)$. (b) *$v_{\mathfrak{P}}(n) = p - 1$ is needed.* $F = p - 2J \in \mathbb{Z}[C_p]$ has $FF^* = p^2$ and $F(\zeta)/F(1) = -1$. (c) *Relation to [LSZ23, Lemma 3.11].* If the coefficients of F lie in a ring unramified at p , then $v_{\mathfrak{P}}(F(1)) \in \{0, p - 1\}$ and the congruence $F(\zeta^j) \equiv F(1) \pmod{y}$ suffices; this is the situation of [LSZ23, Lemma 3.11]. In Theorem 4.1 the coefficients after contraction are sums of elements of μ_p , and Example 4.9 shows that intermediate valuations occur. Re-encoding through the RDS in $\mathbb{Z}[N \times C_p \times C_m]$ to avoid ζ_p does not help: the relevant augmentation is then the value of a character trivial on N , which is 0. The two lemmas do not imply each other: Lemma A allows coefficients involving ζ_p and adds the residue statement, but it needs the group-ring identity $FV = n$, i.e. the norm equation at all p characters of C_p , whereas [LSZ23, Lemma 3.11] only needs the two absolute values $|X|^2 = |Y|^2 = pw$. (d) *Relation to [Fen09].* Feng [Fen09, Lemmas 2, 3] expands $T \in \mathbb{F}_p[E \times \langle a \rangle]$, $p \nmid |E|$, in powers of $a - 1$, using that $\mathbb{F}_p[E]$ is semisimple, and on p. 99 uses that $\binom{i+k}{k}$ is a polynomial of degree k in the exponent i to bound the degree of the fibre functions. Lemma A uses the same device. What is new is its use over the ramified ring $W[\zeta_p]$, with j as the variable and counting over all p points of $\mathbb{F}_p$, and the conclusion that the product $c_\alpha d_\beta$ is constant, which forces residue 1. (e) *Checks.* Complete enumerations of all F with $FF^* = n$ up to symmetry, for (p, d, n) in $\{(5, 4, 5), (5, 3, 5), (5, 3, 15), (5, 3, 20), (5, 6, 30), (5, 8, 5), (5, 8, 10), (7, 4, 7), (7, 4, 14), (11, 1, 11)\}$, show no violation [Comp].

4.3. A ramified alternating product. Let $p \neq q$ be primes, $e \geq 1$, and $G' = C_p \times C_{q^e}$. Call $n \in \mathbb{Z}_{>0}$ *admissible* if (A1) $v_p(n) = 1$ and $v_q(n) \leq e$, and (A2) every prime $r \mid n$ with $r \nmid pq$ is self-conjugate modulo pq . For $x \in \mathbb{Z}[\zeta_p][G']$, $\theta \in \mu_p$ and $\rho \in \mu_q$ (viewed as characters of C_p and of C_{q^e}) write $x(\theta, \rho)$ for the character value.

Theorem 4.12. *Let $x \in \mathbb{Z}[\zeta_p][G']$ satisfy $xx^* = n$ with n admissible. Then*

$$x(1, 1)x(\theta, \rho) = x(\theta, 1)x(1, \rho) \quad \text{for all } \theta \in \mu_p, \rho \in \mu_q.$$

[Lean, for $x = X_a$ with entries in μ_p and $n = pq^e$, inside the proof of Theorem 4.1]

Proof. The claim is trivial if $\theta = 1$ or $\rho = 1$; let $\theta \neq 1 \neq \rho$. All character values are nonzero since $|x(\cdot, \cdot)|^2 = n$. Put

$$\Delta = \frac{x(\theta, \rho)x(1, 1)}{x(\theta, 1)x(1, \rho)} \in \mathbb{Q}(\zeta_{pq}), \quad B = \mathbb{Z}[\zeta_{pq}].$$

We show that Δ is a unit of $B_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ of B , with residue 1 if $\mathfrak{m}$ lies above p or q .

Primes above p (ramified). Let $\mathfrak{m} \mid p$ and $\rho' \in \{\rho, 1\}$. The partial evaluation of C_{q^e} at ρ' gives $F \in \mathbb{Z}[\zeta_p, \zeta_q][C_p]$ with $FF^* = n$. Here $\mathcal{O} = B$ ($d = q$) and $v_{\mathfrak{m}}(n) = (p - 1)v_p(n) = p - 1$ by (A1). By Lemma 4.10, $x(\theta, \rho')/x(1, \rho') = F(\theta)/F(1)$ is a unit of $B_{\mathfrak{m}}$ with residue 1; taking the quotient of the cases $\rho' = \rho$ and $\rho' = 1$, so is Δ .

Primes above q (unramified, with descent). Let $\mathfrak{m} \mid q$ and $\theta' \in \{\theta, 1\}$. Let A be the localization of $A_0 = \mathbb{Z}[\zeta_p]$ at $\mathfrak{m} \cap A_0$; q is unramified in A_0 . The partial evaluation of C_p at θ' gives $F, F^* \in A_0[C_{q^e}]$ with $FF^* = n = q^f b$, $f = v_q(n) \leq e$ by (A1), $b \in A^\times$. By Lemma 4.7, $x(\theta', \rho)/x(\theta', 1) = F(\rho)/F(1)$ is a unit of $A[\rho] \subseteq B_{\mathfrak{m}}$ with residue 1; taking the quotient of the cases $\theta' = \theta$ and $\theta' = 1$, so is Δ .

Other primes. Let $\mathfrak{m}$ lie above $r \nmid pq$. If $r \nmid n$, all four values divide n and are units at $\mathfrak{m}$. If $r \mid n$, then by (A2) and Lemma 4.5 (with $L = pq$) every prime of B above r is self-conjugate, so each of the four values α has $v_{\mathfrak{m}}(\alpha) = \frac{1}{2}v_{\mathfrak{m}}(n)$, and $v_{\mathfrak{m}}(\Delta) = 0$.

Conclusion. Hence $\Delta \in \bigcap_{\mathfrak{m}} B_{\mathfrak{m}} = B$. Every embedding of $\mathbb{Q}(\zeta_{pq})$ commutes with complex conjugation, so Δ has absolute value 1 under every embedding and is a root of unity (Lemma 4.8(i)). It has residue 1 at a prime above p and at a prime above q , so by Lemma 4.8(ii) its order is a power of p and a power of q : $\Delta = 1$. $\square$

Remark 4.13. (a) *Both sides are needed.* Lemma A alone only shows that Δ is a root of unity of p -power order: since $\zeta_p \equiv 1$ modulo every prime above p , the ramified side cannot see $\Delta = \zeta_p^k$. It is the congruence at q that excludes $\zeta_p^k \neq 1$. (b) *Admissibility is needed.* For (A2): with $(p, q, e) = (3, 2, 2)$ and $n = 12 \cdot 49$ (7 is not self-conjugate modulo 6), in a sample of 20,000 of the 38,263,752 elements x with $xx^* = n$, $\Delta \neq 1$ in 30,616 of 40,000 checks, and Δ need not be integral (e.g. $\frac{5}{7} - \frac{8}{7}\zeta_3^2$); with $n = 12 \cdot 25$ (5 is self-conjugate modulo 6) all 52,488 elements have $\Delta = 1$ [Comp]. For (A1): Feng's (56, 7, 56, 8)-RDS [Fen09, Thm. 9] gives a perfect μ_7 -array on $C_7 \times C_2^3$; its contraction to $C_7 \times C_2$ has norm 56, with $v_2(56) = 3 > e = 1$, and $\Delta \neq 1$ in 24 of 42 cases [Comp]. The condition $v_p(n) = 1$ is needed by Remark 4.11(b).

4.4. Proof of Theorem 4.1. Let a be perfect on $C_{pq^e u} \cong C_p \times C_{q^e} \times C_u$ with values in μ_p , and let $x = \varpi_u(X_a) \in \mathbb{Z}[\zeta_p][C_p \times C_{q^e}]$, so that $xx^* = pq^e u$; this n is admissible. Write $x = \sum_{a' \in \mathbb{Z}/p} Z^{a'} H_{a'}$ with $C_p = \langle Z \rangle$ and $H_{a'} \in \mathbb{Z}[\zeta_p][C_{q^e}]$; each coefficient of $H_{a'}$ is a sum of u elements of μ_p .

Step 1: proportional projections. For $j \in \mathbb{Z}/p$ let $g_j = \sum_{a'} \zeta_p^{ja'} H_{a'}$ (the partial evaluation $Z \mapsto \zeta_p^j$). Theorem 4.12 with $\theta = \zeta_p^j$ gives $g_0(1)g_j(\rho) = g_j(1)g_0(\rho)$ for all $\rho \in \mu_q$. Since $|g_0(1)|^2 = pq^e u \neq 0$, Fourier inversion on C_q gives $\varpi_q(g_j) = \kappa_j \varpi_q(g_0)$ with $\kappa_j = g_j(1)/g_0(1)$.

Step 2: a contradiction modulo $\lambda = 1 - \zeta_p$. By Fourier inversion on C_p , $H_0 = \frac{1}{p} \sum_j g_j$, so $E := \varpi_q(H_0) = c_0 \varpi_q(g_0)$ with $c_0 = \frac{1}{p} \sum_j \kappa_j$. Hence $EE^* = c_0 \overline{c_0} \varpi_q(g_0 g_0^*) = c_0 \overline{c_0} pq^e u$ is a scalar. Each coefficient of $E \in \mathbb{Z}[\zeta_p][C_q]$ is a sum of $q^{e-1}u$ elements of μ_p , hence $\equiv q^{e-1}u \pmod{\lambda}$. Complex conjugation induces the identity on $\mathbb{Z}[\zeta_p]/\lambda = \mathbb{F}_p$, so in $\mathbb{F}_p[C_q]$ we get $\tilde{E}\tilde{E}^* = q^{2e-2}u^2 J_q J_q^* = q^{2e-1}u^2 J_q$, with J_q the sum of the elements of C_q ; it has a nonzero coefficient at every element of C_q , as $p \nmid qu$. But the reduction of the scalar EE^* is a scalar. $\square$

Corollary 4.14. *For primes $p \neq q$ and $e \geq 1$ there is no abelian (pq^e, p, pq^e, q^e) -RDS in a group whose Sylow q -subgroup is cyclic.*

Proof. By Proposition 4.4 the group is $C_p \times C_p \times Q$ with $|Q| = q^e$; the forbidden subgroup, of order p , is a direct factor, and Proposition 4.3 gives a perfect μ_p -map on $C_p \times Q \cong C_{pq^e}$ when Q is cyclic. $\square$

Remark 4.15. (a) *Cyclicity is needed.* Feng [Fen09, Thm. 9] constructs a (56, 7, 56, 8)-RDS in $C_7^2 \times C_2^3$, i.e. a perfect μ_7 -array on $C_7 \times C_2^3$; an exhaustive search finds 11,854,080 such arrays and none on $C_7 \times C_8$ [Comp]. (b) *The proof fails at the right place on existing objects.* For the 400 perfect μ_{10} -sequences of length 20, Lemma A, $\Delta = 1$ and the proportionality of Step 1 all hold; Step 2 fails because the alphabet is $\mu_{10} \neq \mu_5$ and the coefficients are not $\equiv q^{e-1}u$ modulo λ [Comp]. (c) *When p^2 divides the length.* The analogue of Lemma A at $v_{\mathfrak{p}}(n) = 2(p-1)$ is

false: for $p = 5$, $\mathcal{O} = \mathbb{Z}[\zeta_{20}]$ and $FF^* = 25$, the valuation of $F(\zeta^j)$ is non-constant for 2,366,200 of the 2,592,720 elements (up to normalization), and ratio residues of Gauss-sum type occur [Comp]. We do not know a replacement strong enough to treat $\text{BH}(p^2q^e, p)$.

4.5. Earlier results on this family. In the alphabet μ_p and with cofactor q^e (in [LSZ23] the letters are interchanged), $f = \text{ord}_p(q)$, the following earlier criteria apply to $\text{BH}(pq^e, p)$, p odd, $e \geq 2$: [DD19a, Thm. 3.3] (q self-conjugate modulo p ; this contains [LSZ23, Thms. 5.1(a), 5.3] for these parameters, the result of Ma and Ng [MN09] as quoted in [Lv17, Sect. 1], and Brock's condition [Bro88] as stated in [DD19a, Result 1.4]); [DD19a, Thms. 3.7, 3.11], the latter for odd e ($f \leq q^e$ or $p \leq (f^2 - q^e)/(f - q^e)$, and $p \leq q^{2e} + q^e + 1$, which is also [LSZ23, Thm. 5.1(b)]); [LSZ23, Thm. 5.4] for even e ; q^e a square modulo p [Fen09, Prop. 8]; $q^e \equiv 1 \pmod{5}$ if $p = 5$ [Fen09, Prop. 5]; $q^e \neq 4$ [FX08]; the results of Liu and Feng [LF16] and of Lv [Lv17, Thm. 1.5], as stated in [Lv17, Sect. 1] ([Lv17, Thm. 1.3] needs q to divide the length exactly once); and Schmidt's field-descent bound [Sch99, LS05] in the form (C4) of Appendix A with $h = p$, with both descent exponents. The count of 34 new lengths above is relative to these criteria; the field-descent bound alone excludes, e.g., 147, 507, 605, 1029, 1083, and [DD19a, Thm. 3.11(ii)] excludes 376 and 568. For $(56, 7)$, $p = 7$ is self-conjugate modulo 8, and Feng's structure theorem [Fen09, Thm. 7] applies, but it does not exclude $E = C_8$. For $(99, 11)$ the condition of [LSZ23, Thm. 5.4(ii)] reads $5 > 5$. For $(1400, 7)$, [DD19a, Thm. 3.11] and [LSZ23, Thm. 5.1(b)] give only $7 \leq 200^2 + 200 + 1$ and $7 \leq 8^2 + 8 + 1$, [LSZ23, Thm. 5.3] (with $q = 5$, $m' = 1$, $n' = 7$) gives only $35 \leq 206$, [LSZ23, Thm. 5.4] needs 200 to be a square, and $200 \equiv 2^2 \pmod{7}$ [Comp]. In Yayla's table [Yay16], $(56, 7)$ and $(99, 11)$ were the two undecided pairs not covered by Theorem 1.1, [FX08] or [LSZ23, Thm. 6.5].

5. RELATED WORK AND STATUS OF THE LITERATURE CHECK

The nearest predecessor. Leung, Schmidt and Zhang [LSZ23, Lemma 3.11] prove: if $X = \sum_{i=0}^{p-1} x_i \zeta_p^i$ and $Y = \sum_i x_i$ with $x_i \in \mathbb{Z}[\zeta_m]$, $p \nmid m$, $p \nmid Y$ and $|X|^2 = |Y|^2 = pw$, $p \nmid w$, then $p \mid X\bar{Y}$. In our notation Y is the augmentation of X_χ and X another of its character values, so the lemma is a statement at the level of character values. The proof of Theorem 1.1 needs two further steps: the conclusion is pushed from the augmentation to all *coefficients* of X_χ (Lemma 2.1), and Parseval's identity is applied to a *single* coset of C_p . The column step is a local form of Ma's lemma [Ma85], [Pot95, Lemma 1.5.1], [DD19a, Result 2.4]; there the input is divisibility of the character values of all of G by a power of p , here it is the norm equation for the partial evaluation in $\mathcal{O}[C_p]$.

Other work. Besides the works cited in Section 3: Brock [Bro88] constrains general $\text{BH}(n, h)$; for perfect p -ary sequences and RDS see also Feng [Fen09], Liu and Feng [LF16], and Zhang and Ge [ZG18] (RDS with $\gcd(m, n) = 1$). Leung and Wang [LW20] and Leung, Li and Mao [LLM23] prove nonexistence of generalized bent functions $\mathbb{Z}_2^n \rightarrow \mathbb{Z}_m$ by vanishing sums; there $\nu_2(|G|) = n$, so Theorem 1.1 applies only for $n = 1$. Ong and Schmidt [OS26] construct $\text{BH}(\mathbb{F}_q \times \mathbb{F}_q, \text{lcm}(6, d))$ from Gauss sums; the Sylow subgroups are not cyclic there, consistent with Remark 2.4.

Leung–Chue–Zhao. Leung, Chue and Zhao [LCZ25] apply vanishing sums to cyclic Butson matrices. According to its abstract and to [OS26, Sect. 1], they prove that every $\text{BH}(\mathbb{Z}_p, h)$, p prime, is monomially equivalent to a Fourier matrix (this covers the case $H = 1$ of Theorem 1.1, cf. Remark 2.5); their abstract also states that they classify the cyclic Butson matrices of order 6 and give a necessary condition for cyclic Butson matrices of order $2p$. These results may contain the cases $n = 6$ and $n = 2p$ of Corollary 1.2, and the 474 pairs with $n = 2p$ (17 with $n = 6$) among the 2063 pairs of Do Duc's list in Section 3 may be excluded there.⁴

⁴Pending: we have not been able to consult the full text of [LCZ25]; the comparison with it is based on its abstract only and will be completed before publication.

Status of the literature check. We have consulted the full texts (in the versions available to us) of [DD19a, DD19b, DDS19, LSZ23, Lv17, Fen09, Yay16, WYZ18, Ara11, HS16, LS05, LS12, LS19, LYZ26, OS26, MS95] and of the arXiv version of [Sch16], and found no statement of Theorem 1.1, of its cyclic case, of Corollary 1.2, or of Theorem 4.1 beyond the cases discussed in Section 4.5. We have *not* been able to consult the full texts of [AdLM02, Ma96, MN09, LCZ25, Pot95, Sch02, dLF11, Mow95] or of [Pei93, Ike99, Fen01, LMF02, FL03b]; every statement we make about these works is taken from the source indicated at the place where it is used, and the comparison with them is provisional.

6. FORMAL VERIFICATION AND COMPUTATIONS

6.1. Lean formalization. The formalization uses Lean 4 (toolchain v4.34.1) and Mathlib (revision d13f23b). It is placed in the Lean library that accompanies [Ope26], but the modules for Theorem 1.1 and Corollary 3.1 depend only on Mathlib. For every theorem in Table 1, `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]`, and the source files contain no `sorry`, `admit`, `native_decide` or `new axiom`. The modules `TheoremS/` (Cyclotomic, Core, LocalRing, Product, Decomposition, Main, Sharp; 784 lines) follow the proof of Section 2. Corollary 3.1 (126 lines) describes D by a homomorphism $\pi : G \rightarrow Q$ with kernel N and a section σ such that every $g \notin N$ is a difference $\sigma(q_1) - \sigma(q_2)$ exactly λ times, and expresses “ N is not a p -group” by a nonzero $x \in \ker \pi$ with $p \nmid \text{ord } x$; it is in `OAI/LinearAlgebra/CirculantHadamard/TheoremS/RDS.lean`.

The modules `RamifiedComparison/` (Core, Localized, PrimeCyclotomic, LemmaA, Auxiliary, TheoremC, PrimePower; 2338 lines; namespace `OAI.CirculantHadamard.RamifiedComparison`) formalize Section 4. The main statements are:

```
theorem lemmaA (p d : ℕ) [hp : Fact p.Prime] (hd : ¬ p ∣ d)
  [IsCyclotomicExtension {p * d} ℚ K] {ζ : K} (hζ : IsPrimitiveRoot ζ p)
  (P : Ideal (ℳ K)) [hP : P.IsPrime] [hPo : P.LiesOver (Ideal.span {(p : ℤ)})]
  (u v : ℤMod p → ℳ K) (n : ℤ) (hn1 : (p : ℤ) ∣ n) (hn2 : ¬ (p : ℤ) ^ 2 ∣ n)
  (huv : ∀ τ : ℤMod p, ∑ a, u a * v (τ - a) = if τ = 0 then (n : ℳ K) else 0) :
  ∃ α : ℕ, ∀ j : ℕ,
    ∑ a, u a * hζ.toInteger ^ (j * a.val) ∈ P ^ α ∧
    ∑ a, u a * hζ.toInteger ^ (j * a.val) ∉ P ^ (α + 1) ∧
    (∑ a, u a * hζ.toInteger ^ (j * a.val)) - ∑ a, u a ∈ P ^ (α + 1)

theorem not_exists_perfect_prime_power_phase (p q : ℕ) [hp : Fact p.Prime]
  [hq : Fact q.Prime] (hpq : p ≠ q) (e : ℕ) (he : 1 ≤ e) :
  ¬ ∃ a : ℤMod (p * q ^ e) → ℂ,
    (∀ j, a j ^ p = 1) ∧
    ∀ τ : ℤMod (p * q ^ e), τ ≠ 0 →
      ∑ j, a j * (starRingEnd ℂ) (a (j + τ)) = 0
```

`lemmaA` is Lemma 4.10 for $n \in \mathbb{Z}$ with $v_p(n) = 1$ (not for general $n \in \mathcal{O}$ with $v_{\mathfrak{p}}(n) = p - 1$), for an arbitrary second factor V and an arbitrary prime $\mathfrak{P}$ above p . `not_exists_perfect_prime_power_phase` is Theorem 4.1(i) for all primes $p \neq q$, including $p = 2$ and $q = 2$. Theorem 4.12 is formalized only inside this proof, for $x = X_a$ with coefficients $a(g) = \zeta_p^{k(g)}$ (hypothesis `hu`) and $n = pq^e$. At the primes above q the formal proof uses the comparison lemma of the Lean library of [Ope26] (`actual_local_character_association`); Lemma 4.7 as stated, with the descent for $f \geq 2$, is not formalized.

6.2. Computations. No proof depends on a computation. The pairs of Do Duc’s list were transcribed from the list referenced in [DD19a, Remark 3.13]; the counts 2262, 61, 364 were obtained by two independent scripts, and the comparison with earlier criteria by a third. The searches of Remark 2.4 and Example 3.2 enumerate all $f : G \rightarrow \mu_h$ with $f(0) = 1$ (up to $8.6 \cdot 10^9$

TABLE 1. Lean theorems (namespace `OAI.CirculantHadamard.TheoremS`, resp. `OAI.CirculantHadamard.RamifiedComparison` for the last two rows of the first part) and the statements they formalize.

<i>Formally verified statements</i>	
<code>no_perfect_function_sharp</code>	Theorem 1.1
<code>no_perfect_sequence_sharp</code>	Theorem 1.1 for $G = \mathbb{Z}/n$
<code>no_perfect_function, no_perfect_sequence</code>	Theorem 1.1 for $p \nmid h$; Corollary 3.3(i), and (ii) for odd h
<code>no_perfect_sequence_two_mod_four</code>	Corollary 3.3(ii) for $G = \mathbb{Z}/n$ and $h \equiv 2 \pmod{4}$
<code>no_perfect_quaternary_sequence</code>	the case $h = 4$, p odd
<code>RDSCheck.no_semiregular_rds</code>	Corollary 3.1
<code>lemmaA</code>	Lemma 4.10 for $n \in \mathbb{Z}$
<code>not_exists_perfect_prime_power_phase</code>	Theorem 4.1(i); Theorem 4.12 for $x = X_a$, $n = pq^e$
<i>Derived from the above by arguments that are not formalized</i>	
Corollary 1.2	uses the constructions of Mow and Duc–Schmidt
Corollary 3.4	translation to generalized bent functions
Corollary 3.5	eleven instances of <code>no_perfect_quaternary_sequence</code> ; the second sentence relies on Arasu’s remark
<i>Not formalized:</i> counts for Do Duc’s list, Lemma 3.6 and its applications, Corollary A.1, Lemma 4.7, Theorem 4.12 for general admissible n , Theorem 4.1(ii), Corollaries 4.2, 4.14, the counts of Section 4	

functions), checking all autocorrelations in floating point, or use a meet-in-the-middle search for $G = K \times C_2$ based on $|\chi(c_0)|^2 + |\chi(c_1)|^2 = |G|$ and $\operatorname{Re}(\chi(c_0)\overline{\chi(c_1)}) = 0$; the two methods agree on overlapping cases. Corollary A.1 was computed by a criterion engine and reproduced by an independent implementation; the descent exponents of [LS05, Def. 2.6] and [LS12, Def. 2.10] agree on all 33,223 pairs (L, N) used, and both reproduce $F(165^2, 165) = 3 \cdot 5^2 \cdot 11^2$ [LS05, Example 2.7].

For Section 4: complete enumerations for Lemma 4.10 (Remark 4.11(e)); for Theorem 4.12, all 1944, 504 and 52,488 elements for $(p, q, e, n) = (3, 2, 2, 12)$, $(3, 7, 1, 21)$, $(3, 2, 2, 300)$, and the negative controls of Remark 4.13(b); the chain of the proof on the 400 perfect μ_{10} -sequences of length 20 (Example 4.9, Remark 4.15(b)); the search for perfect μ_7 -arrays on $C_7 \times C_8$ and $C_7 \times C_2^3$ (Remark 4.15(a)); the valuation statistics of Remark 4.15(c); and the classification of the lengths $pq^e \leq 2000$ against the earlier criteria of Section 4.5, together with the check of $(63, 3)$ against the criteria of [DD19a] (Section 3.6), by a stdlib-only Python script. These enumerations were written independently of the programs that first produced the corresponding claims.

6.3. Artifact and data availability. The Lean development is on the branch `contrib/general-norm-extensions` of our fork of the Lean library of [Ope26], based on upstream commit `fd4aeeb2ee4f`; the `TheoremS` modules will be committed there at release (the `RamifiedComparison` modules are contained in commit `08fcd618213a`), and the commit hash, `lake-manifest.json`, build log and `#print axioms` output will be published with the artifact, together with the computation scripts and logs and the transcription of Do Duc’s list with its SHA-256 checksum.

USE OF AI TOOLS

Parts of this work were carried out with the assistance of AI systems, which were used to explore arguments, to draft and revise parts of the text, to write computer programs for the computations, and to write parts of the Lean formalization. All mathematical statements and

proofs in this paper have been checked by the authors. The statements listed in the first part of Table 1 have been verified in Lean 4, with the scope indicated there, and the numerical results have been reproduced by independent programs. The authors take full responsibility for the content of the paper.

APPENDIX A. PERFECT QUATERNARY SEQUENCES OF LENGTH $N \equiv 0 \pmod{4}$

Corollary A.1. *Let $N \equiv 0 \pmod{4}$, $N > 16$. Exclude N if one of the following criteria applies: (C1) N is not a sum of two squares [Ara11, Thm. 4.5]; (C2) $N = 2^a$, $a \geq 5$ [DDS19, Thm. 3.6]; (C3) an odd prime divisor of N is self-conjugate modulo N [DD19a, Thm. 3.3] (with $h = 4$); (C4) Schmidt's bound $N \leq C^2 F^2 / (4\varphi(F))$, for a character of any order $d \mid N$, $L = \text{lcm}(4, d)$, $C = 4N/L$ and $F = F(L, N)$ [Sch99, LS05, LS12], is violated; (C5) Lemma 3.6; (C6) Theorem 1.1; (C7) for $N \leq 1000$, [Ara11, Remark after Thm. 4.11] together with Corollary 3.5. Then the lengths not excluded by these criteria are [Comp]*

- up to 5000: $3528 = 2^3 \cdot 3^2 \cdot 7^2$ and $4900 = 2^2 \cdot 5^2 \cdot 7^2$;
- up to 10^5 : in addition 7056, 8712, 9800, 17424, 19600, 34848, 39200, 44100, 88200 (11 in total);
- up to 10^6 : in addition 108900, 176400, 213444, 217800, 352800, 426888, 435600, 544500, 705600, 736164, 853776, 871200 (23 in total).

We did not implement every lemma of [AdLM02], so for $N > 1000$ these lists are upper bounds for the set of open lengths. Without (C7), (C1)–(C6) leave in addition 72, 100, 144, 784. Up to 10^5 , (C1)–(C5) exclude 19,382 of the 25,000 lengths, Theorem 1.1 excludes 5,604 of the others, 3 exist and 11 remain. Table 2 lists the lengths in $(1000, 10^6]$ excluded by (C5) alone; below 1000, (C5) alone excludes 200, 288, 900, which are also excluded according to [Ara11]. Odd $N > 1$ are impossible, and lengths $N \equiv 2 \pmod{4}$ with an odd prime of exponent 1 are excluded by Theorem 1.1; the other lengths $N \equiv 2 \pmod{4}$ are not considered here.

TABLE 2. Lengths $N \in (1000, 10^6]$, $N \equiv 0 \pmod{4}$, excluded by Lemma 3.6 and by no other criterion of Corollary A.1; here $d = L = N$, $C = 4$, and $F = F(N, N)$ is the same for [LS05, Def. 2.6] and [LS12, Def. 2.10].

N	factorization	$F(N, N)$	$4F^2/\varphi(F)$
1800	$2^3 \cdot 3^2 \cdot 5^2$	$120 = 2^3 \cdot 3 \cdot 5$	1800
3600	$2^4 \cdot 3^2 \cdot 5^2$	$240 = 2^4 \cdot 3 \cdot 5$	3600
6084	$2^2 \cdot 3^2 \cdot 13^2$	$468 = 2^2 \cdot 3^2 \cdot 13$	6084
12100	$2^2 \cdot 5^2 \cdot 11^2$	$1100 = 2^2 \cdot 5^2 \cdot 11$	12100
12168	$2^3 \cdot 3^2 \cdot 13^2$	$936 = 2^3 \cdot 3^2 \cdot 13$	12168
14112	$2^5 \cdot 3^2 \cdot 7^2$	$1008 = 2^4 \cdot 3^2 \cdot 7$	14112
24200	$2^3 \cdot 5^2 \cdot 11^2$	$2200 = 2^3 \cdot 5^2 \cdot 11$	24200

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